

1. INTRODUCTION

Road accidents are one of the biggest problems in today's world. Every day, many people lose their lives due to accidents on highways and city roads. In most cases, the main reason for deaths is the delay in providing medical help after the accident occurs. Sometimes people nearby may not notice the accident immediately, especially during night time or in remote areas. Because of this delay, injured people may not receive treatment at the right time.

To solve this problem, an automatic accident detection and alert system is required. The "Accident Detection and Alert System using PIC Microcontroller" is developed to provide quick emergency support by automatically detecting accidents and sending alert messages to emergency contacts. The system helps in reducing rescue time and improving road safety.

In this project, a PIC microcontroller is used as the main controlling unit. It continuously monitors the vehicle condition using sensors such as a vibration sensor. These sensors detect sudden shocks, impacts, or abnormal vibrations that occur during an accident. When the detected value crosses a predefined limit, the PIC microcontroller identifies it as an accident condition.

After detecting the accident, the system immediately activates the GSM module to send an emergency SMS alert. At the same time, the GPS module collects the exact location coordinates of the accident place. The location details are included in the alert message so that family members, ambulance services, or rescue teams can quickly reach the accident location.

An LCD display is also connected to the system to show the current status and alert messages. A buzzer is used to provide an audio warning indication during emergency conditions. The PIC microcontroller is selected because it is reliable, easy to program, and consumes less power. It can easily interface with sensors

and communication modules, making it suitable for embedded safety applications.

This system is useful for cars, bikes, buses, and other vehicles. It provides automatic monitoring without human involvement and works continuously in real time. The project is compact, economical, and easy to install in vehicles.

Overall, the “Accident Detection and Alert System using PIC Microcontroller” is an effective solution for improving vehicle safety and reducing the response time during emergencies. By using embedded systems and wireless communication technologies, the project helps in saving human lives and supporting smart transportation systems.

1.1 Background and Context:

In recent years, the number of vehicles on roads has increased rapidly due to urbanization, industrial growth, and improved transportation facilities. Along with this growth, the rate of road accidents has also increased significantly. According to various road safety reports, thousands of people lose their lives every year because emergency medical assistance does not reach the accident location on time. In many situations, the severity of injuries increases due to delays in accident reporting and rescue operations. Traditional accident reporting systems mainly depend on eyewitnesses, nearby people, or drivers to inform hospitals, police stations, or family members. However, during night time, on highways, or in remote areas, accidents may remain unnoticed for a long time. In such cases, the delay in communication becomes a major reason for loss of human life. Therefore, there is a strong need for an intelligent and automatic accident detection system that can quickly report accidents without requiring human involvement. With the advancement of embedded systems and wireless

communication technologies, smart safety systems can now be developed for vehicles. Technologies such as sensors, microcontrollers, GSM, and GPS make it possible to automatically detect accidents and send emergency alerts in real time. Embedded systems provide fast processing, compact size, low power consumption, and reliable performance, making them suitable for vehicle safety applications. The proposed “Accident Detection and Alert System using PIC Microcontroller” is developed to address these challenges. In this system, vibration and accelerometer sensors continuously monitor vehicle movement and detect sudden impacts or abnormal conditions during accidents. The PIC microcontroller processes the sensor data and identifies accident situations accurately. Once an accident is detected, the GSM module automatically sends an emergency SMS alert to predefined contacts such as family members, hospitals, or rescue teams. At the same time, the GPS module provides the exact location coordinates of the accident place, helping emergency services reach the victim quickly. This system improves road safety by reducing the delay between accident occurrence and emergency response. It is compact, cost-effective, reliable, and suitable for installation in different types of vehicles. Thus, the project contributes to the development of smart transportation and intelligent vehicle safety systems.

1.2 Relevance and Sustainability Development Aspect:

The project is highly relevant in modern transportation systems where road safety has become a major concern. The system helps reduce fatalities by minimizing the delay in emergency response.

Sustainability aspects of the project include:

- Low power consumption using embedded electronics.
- Reduced dependency on manual accident reporting.

- Compact and portable design.
- Cost-effective implementation suitable for large-scale use.
- Improved public safety and emergency management.

The project contributes to smart transportation and intelligent safety systems. From the sustainability point of view, the project provides many benefits. The system is compact, low-cost, and consumes less power, making it suitable for long-term use in vehicles. It reduces the dependency on manual accident reporting and improves emergency management.

The use of automatic monitoring systems can help save human lives and reduce the impact of road accidents on society. It encourages the use of modern electronics and communication technologies for public safety. Since the system can be installed in different types of vehicles such as cars, bikes, buses, and trucks, it has wide practical applications.

Another important aspect of sustainability is that the project minimizes rescue delays, which can reduce the severity of injuries and improve survival chances. By providing faster communication during emergencies, the system helps medical teams respond quickly and efficiently. Overall, the “Accident Detection and Alert System using PIC Microcontroller” is a useful, practical, and socially beneficial project. It improves road safety, supports emergency response systems, and contributes to the development of smart and sustainable transportation technology.

1.3 Interdisciplinary and Multidisciplinary Aspects:

A) Interdisciplinary Aspects

The interdisciplinary aspect of this project refers to the integration of concepts from different fields to solve a single problem in a connected manner.

- Integration of Embedded Systems and Electronics Engineering using PIC microcontroller and sensors.
- Combination of Wireless Communication and Automobile Safety for sending accident alerts through GSM/GPS modules.
- Use of Programming Skills with Hardware Interfacing for real-time accident detection and monitoring.
- Application of Sensor Technology with Emergency Management Systems to provide quick accident notifications.
- Integration of IoT concepts with vehicle monitoring for smart safety solutions.

B) Multidisciplinary Aspects

The multidisciplinary aspect refers to the involvement of multiple fields working on the same project, but each contributing its own part independently.

- Combination of Electronics and Telecommunication Engineering for sensor interfacing and communication systems.
- Use of Computer Programming for developing embedded C code in the PIC microcontroller.
- Integration of Wireless Communication Technology for transmitting accident alert and location information.

- Application of Safety and Emergency Management concepts for faster rescue and medical response.
- Inclusion of IoT and Smart Monitoring Systems for real-time accident detection and alert generation.

1.4 Scope and Objectives:

A) Scope of the Project

The scope of the project is to develop a smart vehicle safety system that can automatically detect accidents and send alert messages using a PIC microcontroller. The system helps in reducing the delay in emergency response during road accidents. It can be installed in vehicles to improve passenger safety and provide immediate accident information to family members or emergency services.

- Automatic accident detection in vehicles.
- Real-time location tracking.
- SMS alert system for emergency contacts.
- IoT integration for remote monitoring.
- Can be used in cars, bikes, buses, and emergency vehicles.

B) Objectives of the Project

The main objectives of this project are:

- To design and implement a microcontroller-based accident detection system using the 8051 architecture.
- To accurately detect collision events using a vibration sensor interfaced with the controller.
- To acquire real-time geographical coordinates through a GPS receiver module.

- To transmit automated emergency alerts containing location data via a GSM communication module.
- To provide local indication of system status using an LCD display and buzzer.
- To develop a reliable, low-cost, and real-time embedded system aimed at minimizing emergency response time and enhancing road safety.

1.5 Motivation:

Road accidents are increasing rapidly due to heavy traffic, overspeeding, and lack of immediate emergency assistance. In many cases, injured people do not receive medical help on time because the accident information is not communicated quickly to family members or rescue services. This delay in response can lead to serious injuries and loss of lives.

The main motivation behind developing the Accident Detection and Alert System using PIC Microcontroller is to improve vehicle safety and provide immediate accident notifications automatically. The project aims to reduce human effort in reporting accidents and ensure faster emergency response by using sensor technology and wireless communication.

This system is designed to detect accidents in real time and send alert messages with location details to emergency contacts. The project also promotes the use of embedded system.

Overall, the motivation of the project is to contribute towards safer roads, quicker medical assistance, and reduction in accident-related fatalities. Overall, the project focuses on saving human lives by providing quick accident detection and immediate communication with rescue services and family members.

1.6 Problem Statement:

“Road accidents often go unreported for a long time, especially in remote areas, causing delays in medical assistance and increasing the risk of fatalities. To solve this, an automated system using an PIC microcontroller, vibration sensor,

GPS, and GSM module is needed to detect accidents and instantly send the location to emergency contacts, ensuring quick response and improved safety.”

2. LITERATURE SURVEY

Many researchers have developed accident detection and alert systems using embedded systems and IoT technologies to reduce emergency response time. Existing systems mainly use sensors, GPS, GSM, and Wi-Fi modules for accident detection and location tracking.

Several studies used accelerometers and vibration sensors to detect vehicle collisions and automatically send alerts to emergency contacts. GPS modules were commonly used to obtain real-time vehicle location, while communication modules such as GSM and ESP32 were used to transmit notifications through SMS or internet services.

Research shows that automatic accident alert systems help rescue teams reach accident locations faster and improve road safety. IoT-based systems provide better real-time monitoring and communication compared to traditional manual reporting methods.

Based on the literature survey, the proposed system using PIC18F452, SW-420, NEO-6M GPS, and ESP32 offers a simple, low-cost, and efficient solution for automatic accident detection and emergency alert notification.

2.1 Introduction:

Road accidents are increasing rapidly due to heavy traffic, over-speeding, and careless driving. In many situations, accident victims do not receive immediate

medical assistance because the accident information is not reported quickly. Delayed emergency response can lead to serious injuries and loss of life.

To overcome this problem, an automatic accident detection and alert system can be used to detect accidents and immediately send emergency notifications with the vehicle location. The proposed system uses PIC18F452, SW-420, NEO-6M GPS, and ESP32 to develop a smart vehicle safety system. The SW-420 vibration sensor detects sudden impacts or strong vibrations caused by accidents. The PIC18F452 microcontroller processes the sensor data and identifies accident conditions. After detection, the NEO-6M GPS module obtains the current latitude and longitude of the vehicle.

The ESP32 module then sends an alert message containing the accident location to emergency contacts through internet-based communication. This system helps reduce rescue time, improves emergency response efficiency, and increases road safety. The project is low-cost, compact, reliable, and suitable for cars, buses, ambulances, and other vehicles.

2.2 Review of Technical Papers Referred / Field Visit:

Review of Technical Papers Referred

Several technical papers related to accident detection and alert systems were studied during the development of this project. The reviewed papers explained the use of vibration sensors, accelerometers, GPS modules, GSM, and IoT communication technologies for automatic accident detection and emergency notification.

From the literature review, it was observed that:

- Vibration and accelerometer sensors are effective in detecting vehicle collisions.
- GPS modules provide accurate real-time vehicle location.
- IoT and wireless communication modules help send instant emergency alerts.
- Automatic alert systems reduce rescue response time and improve road safety.

The reviewed papers also highlighted the importance of reducing false accident detection and improving communication reliability. Based on these studies, the proposed system uses PIC18F452, SW-420, NEO-6M GPS, and ESP32 to develop a simple, low-cost, and efficient accident detection and alert system.

A field study was conducted to understand the practical challenges involved in accident detection and emergency response systems. During the observation, it was found that delays in informing emergency services are one of the major reasons for increased accident severity.

The study also identified:

- Lack of immediate accident reporting in remote areas
- Delay in sharing exact accident location
- Dependence on manual communication methods

Based on the field observations, the proposed system was designed to automatically detect accidents and send real-time location alerts to emergency contacts using GPS and IoT technology. The system can help improve rescue operations and reduce response time during emergencies

2.3 Conclusion:

From the literature survey, it is observed that accident detection and alert systems play an important role in improving road safety and reducing emergency response time. Many existing systems use vibration sensors, accelerometers, GPS modules, GSM, and IoT technologies for detecting accidents and sending emergency notifications.

The reviewed papers show that automatic accident detection systems can quickly identify vehicle collisions and provide accurate location information to rescue teams or emergency contacts. IoT-based communication methods offer faster and more reliable alert transmission compared to traditional manual reporting methods.

However, some existing systems face limitations such as false accident detection, high cost, and communication delays. To overcome these issues, the proposed system uses PIC18F452, SW-420, NEO-6M GPS, and ESP32 to develop a simple, reliable, and cost-effective accident detection and alert system capable of sending real-time emergency notifications with location details.

3. PROJECT SYSTEM DESIGN METHODOLOGY

3.1 Introduction:

The **Accident Detection and Alert System** is a smart embedded and IoT-based safety solution developed to automatically identify road accidents and immediately inform emergency contacts with precise location details. In today's fast-moving world, road accidents are increasing due to overspeeding, driver

distraction, and unpredictable road conditions. One of the major challenges after an accident is the delay in communication, which often leads to late medical assistance and increased risk of fatalities.

This project aims to overcome this issue by designing an automated system that can detect accidents without human intervention and send instant alerts. The system is built using PIC18F4520 as the central processing unit, a vibration sensor for detecting sudden impacts, a GPS module for tracking real-time location, and ESP32 for transmitting emergency notifications through internet connectivity.

The vibration sensor continuously monitors vehicle movement and detects sudden shocks that may indicate a collision. Once an abnormal vibration is detected, the microcontroller processes the signal and verifies whether it corresponds to an accident condition. After confirmation, the GPS module retrieves the exact latitude and longitude of the vehicle.

The ESP32 module then sends this critical information to emergency contacts or cloud services in the form of an alert message, which may include a Google Maps link for easy location tracking. This ensures that help can be dispatched quickly to the accident site.

The system is designed to be low-cost, compact, and reliable, making it suitable for implementation in various vehicles such as cars, buses, and trucks. It enhances road safety by reducing dependency on manual reporting and enabling faster emergency response.

3.2 Block diagram of system and working of the system:

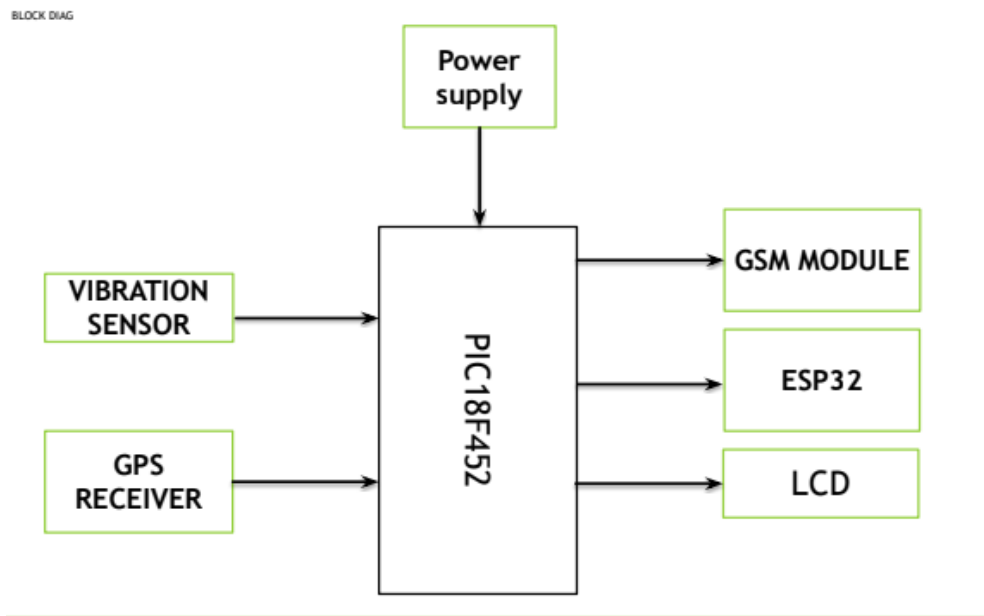
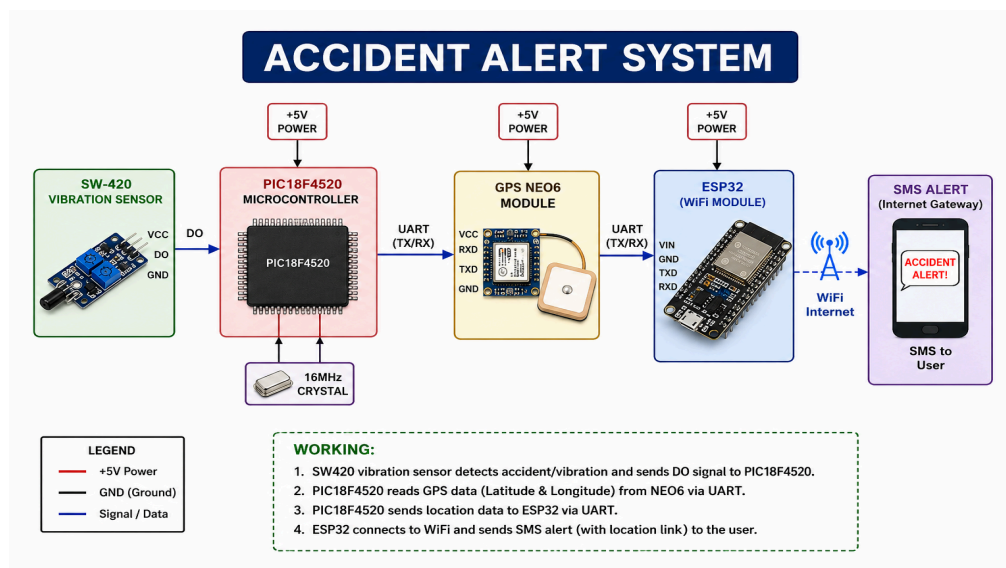


Fig 3.2.1 Block Diagram



Block Diagram Explanation

1. Power Supply Unit

- Provides regulated DC power (typically 5V) to all components.
 - Ensures stable operation of microcontroller, sensor, GPS, and ESP32.
 - Prevents system failure due to voltage fluctuations.
-

2. Vibration Sensor Module

- Uses a vibration sensor to detect sudden shocks or impacts.
 - Continuously monitors vehicle movement.
 - When a strong vibration occurs (possible accident), it generates a digital signal.
 - Acts as the primary accident detection unit.
-

3. PIC18F4520 (Main Controller)

- Central processing unit of the system.
 - Receives input from the vibration sensor.
 - Checks whether the vibration exceeds a predefined threshold.
 - If accident is confirmed, it triggers GPS reading and alert process.
 - Communicates with GPS module and ESP32 via UART.
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4. GPS Module

- Provides real-time location of the vehicle.

- Continuously tracks latitude and longitude.
 - Sends location data to PIC18F4520 in NMEA format.
 - Helps in identifying exact accident location for rescue teams.
-

5. ESP32 Module

- Acts as the communication interface for internet connectivity.
- Receives accident data and GPS coordinates from PIC18F4520.
- Connects to Wi-Fi network.
- Sends emergency alerts to cloud server, mobile app, or messaging service.

Working Summary:

Vibration Sensor + GPS Module → PIC Microcontroller → ESP32 (IoT communication) → User Mobile/Email Alert → Location Display

3.4 Specifications:

1. ESP32

The ESP32 is a powerful and widely used microcontroller designed for IoT applications. It comes with built-in Wi-Fi and Bluetooth capabilities, making it suitable for wireless communication and real-time data transmission.

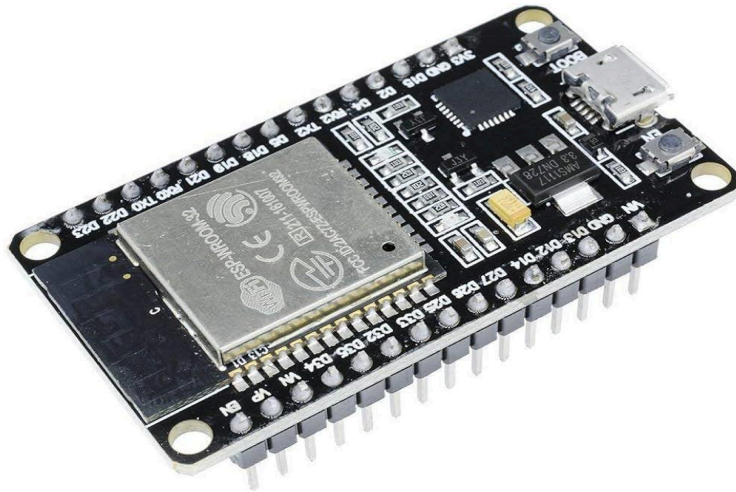


Fig 3.4.1 ESP32

Description:

ESP32 acts as the main controller of the system. It collects data from sensors, processes it, and sends it to cloud platforms like Blynk for remote monitoring. Its high processing speed and wireless connectivity make it ideal for IoT-based health monitoring systems.

Pin description:



Fig 3.4.2 Pin description: (ESP32)

Specifications:

- **Microcontroller Type:** 32-bit Xtensa LX6 Dual-Core Processor
- **Operating Voltage:** 3.0V – 3.6V
- **Clock Frequency:** Up to 240 MHz
- **Wi-Fi:** 802.11 b/g/n (built-in)
- **Bluetooth:** v4.2 (Classic + BLE)
- **RAM:** 520 KB SRAM
- **Flash Memory:** Typically 4 MB
- **GPIO Pins:** Up to 34 programmable pins
- **Communication Protocols:** UART, SPI, I2C, I2S, CAN
- **ADC Channels:** 12-bit SAR ADC (multiple channels)
- **DAC Channels:** 2 (8-bit)
- **Power Consumption:** Low power modes available
- **Operating Temperature:** -40°C to +125°C

2.PIC Universal kit (18F4520)

The **PIC18F4520** is a high-performance 8-bit microcontroller from Microchip Technology, based on the enhanced **PIC18 architecture**. It is widely used in embedded systems due to its low power consumption, rich peripheral set, and ease of interfacing with sensors and communication modules.

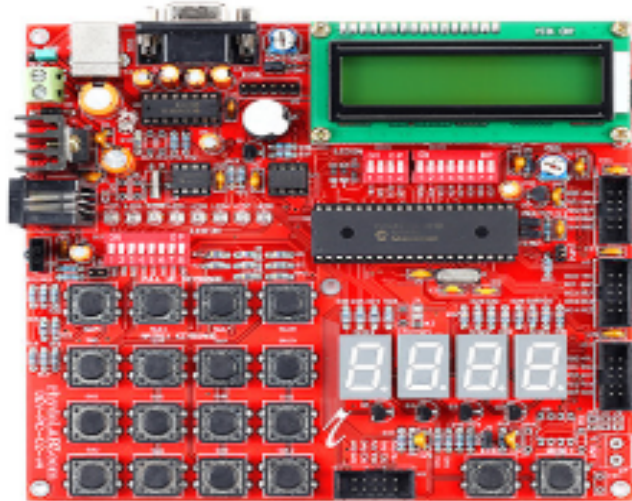


Fig 3.4.3 PIC Universal Kit (18f4520)

Description:

It features a **32 KB Flash program memory**, **1.5 KB RAM**, and **256 bytes EEPROM**, making it suitable for medium-complexity embedded systems. The microcontroller operates at speeds up to **40 MHz** and provides high processing efficiency with up to **10 MIPS performance**.

PIC18F4520 includes a rich set of peripherals such as a **10-bit ADC (13 channels)** for analog signal processing, multiple **timers**, **PWM modules**, and communication interfaces like **USART**, **SPI**, and **I2C**, enabling easy interfacing with sensors and modules such as GPS, vibration sensors, and wireless communication devices.

Pin description:

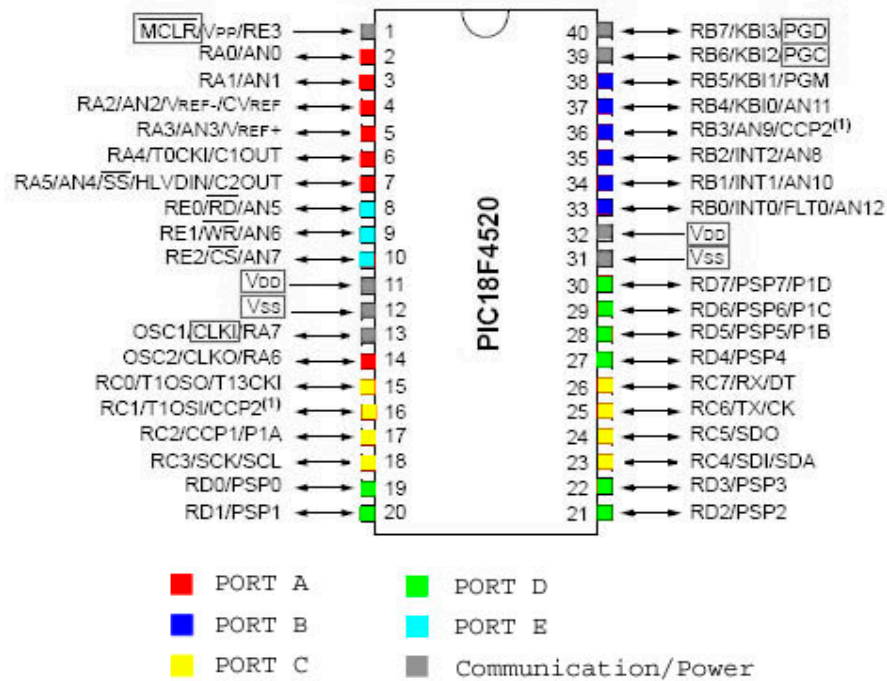


Fig 3.4.4 Pin description: PIC 18F4520

Specifications:

- **Microcontroller Type:** 8-bit PIC18 RISC Microcontroller
- **Core Architecture:** PIC18 Enhanced Harvard Architecture
- **Operating Voltage:** 2.0V – 5.5V
- **Clock Frequency:** Up to 40 MHz
- **Instruction Speed:** Up to 10 MIPS
- **Flash Program Memory:** 32 KB
- **SRAM:** 1536 Bytes
- **EEPROM Data Memory:** 256 Bytes

- **GPIO Pins:** 36 programmable I/O pins
- **Timers:** 4 Timers (8-bit and 16-bit)
- **ADC Resolution:** 10-bit ADC
- **ADC Channels:** 13 Analog Input Channels
- **Communication Protocols:** USART, SPI, I2C
- **CCP/PWM Modules:** 2 CCP (Capture/Compare/PWM) modules
- **Interrupts:** Multiple interrupt sources with priority levels
- **Oscillator Support:** Internal and External Oscillator support
- **Programming Support:** ICSP (In-Circuit Serial Programming)
- **Operating Temperature:** -40°C to +125°C (industrial grade variants)
- **Typical Applications:** Embedded systems, robotics, automation, sensor interfacing

3. NEO-6M GPS Module

The NEO-6M is a widely used GPS receiver module that provides accurate real-time location information such as latitude, longitude, and altitude. It communicates with the microcontroller using serial communication and is suitable for tracking applications.

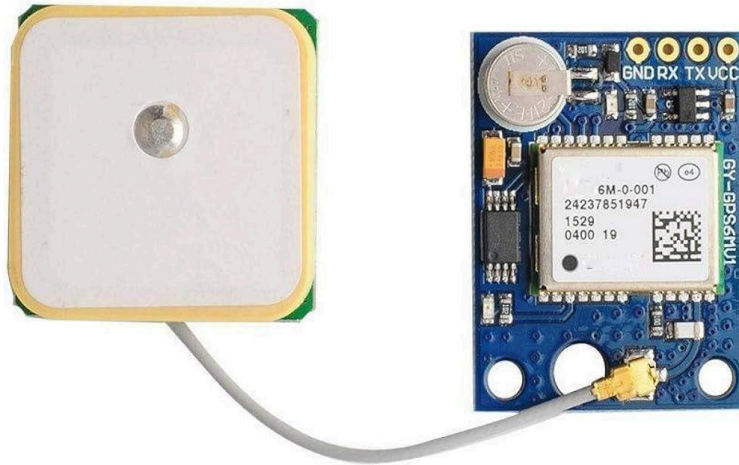
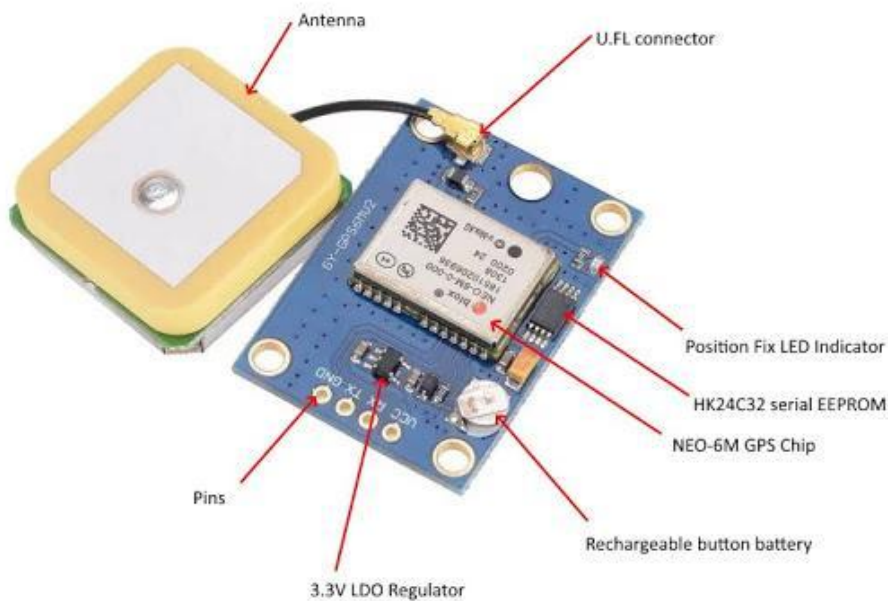


Fig 3.4.6 NEO-6M GPS Module

Description:

The NEO-6M GPS module is used to track the real-time location of the user. It receives signals from satellites and provides location coordinates, which are processed by the ESP32. This information is useful in emergency situations to identify the exact position of the patient and provide quick assistance.

Pin description:

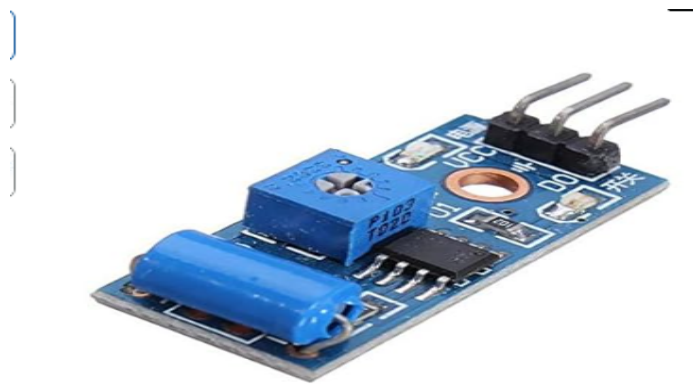


Specifications:

- **Operating Voltage:** 3.3V – 5V
- **Position Accuracy:** Up to 2.5 meters
- **Communication Protocol:** UART (Serial Communication)
- **Update Rate:** 1 Hz (default), up to 5 Hz
- **Cold Start Time:** ~27 seconds (approx.)
- **Hot Start Time:** ~1 second (approx.)
- **Sensitivity:** -161 dBm (tracking)
- **Antenna:** External active antenna
- **Operating Temperature:** -40°C to +85°C

4.Vibration Sensor

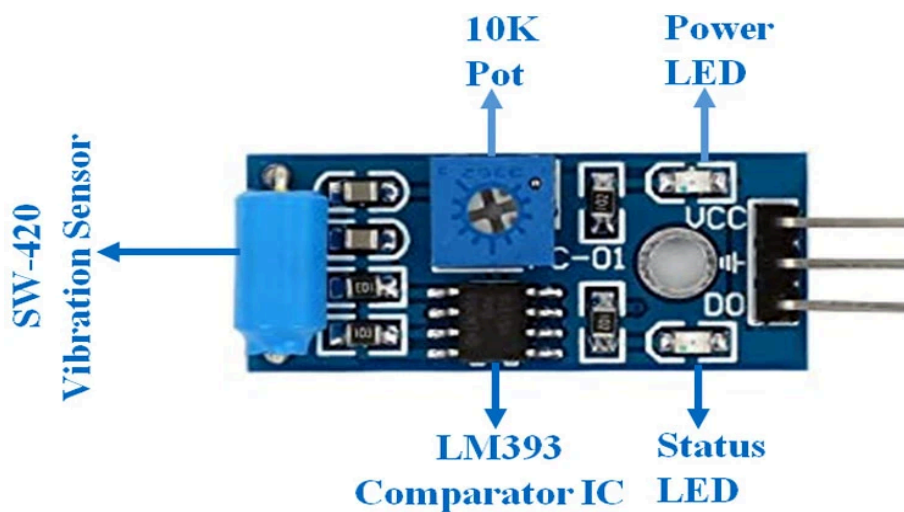
A Vibration Sensor is an electronic device used to detect vibrations, shocks, or movements in machines, objects, or surfaces. It converts mechanical vibration into an electrical signal that can be processed by a microcontroller or monitoring system. Vibration sensors are commonly used in security systems, industrial machine monitoring, robotics, automobiles, and earthquake detection systems.



Description:

When vibration or movement occurs, the sensor produces either a digital HIGH/LOW output or an analog signal depending on the sensor type. These sensors help identify abnormal machine behavior, impacts, or motion.

Pin Description:



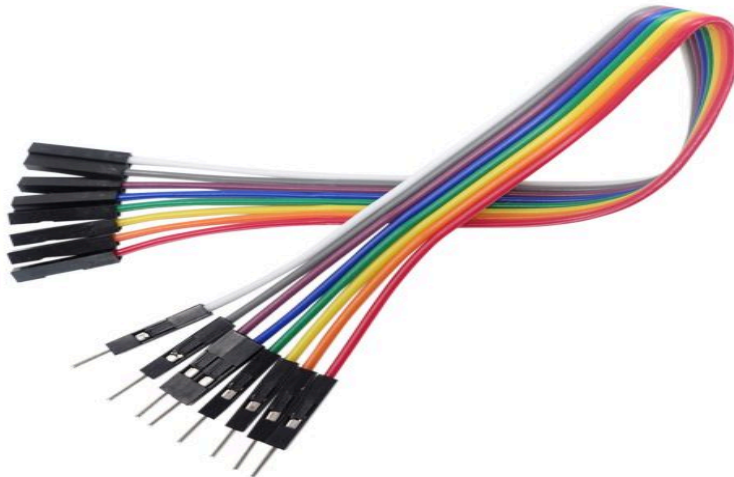
Specifications:

- **Sensor Type:** Vibration / Shock Detection Sensor
- **Operating Voltage:** 3.3V – 5V DC
- **Output Type:** Digital Output / Analog Output (depends on module)
- **Sensitivity:** Adjustable using onboard potentiometer (in modules)
- **Detection Method:** Spring-based or Piezoelectric sensing
- **Response Time:** Fast response to vibration and shock

- **Interface: Compatible with Microcontrollers like Arduino, PIC, ESP32, Raspberry Pi**
- **Power Consumption: Low power consumption**
- **Indicator LED: Power LED and status LED available in modules**
- **Operating Temperature: -10°C to +70°C (typical)**

5. Connecting Wires

Connecting wires are used to establish electrical connections between different components in the circuit.



Description:

Connecting wires are used to link all components such as sensors, microcontroller, display, and power supply. They provide proper signal and power transmission throughout the circuit. Their flexibility and ease of use make them ideal for prototyping and assembling the IoT-based health monitoring system.

Specifications:

- **Type:** Jumper wires (Male-to-Male / Male-to-Female)
- **Material:** Copper conductor with insulation
- **Length:** Standard (10 cm – 30 cm approx.)
- **Current Capacity:** Suitable for low-power circuits
- **Flexibility:** High

4. DEVELOPMENT AND TESTING**4.1 Introduction:**

The development and testing phase of the Accident Detection and Alert System focuses on designing, implementing, and verifying the functionality of the hardware and software modules. The system was developed using the PIC18F4520 microcontroller, vibration sensor, GPS module, and ESP32.

During development, each component was individually tested and later integrated into a complete system. The testing process ensured accurate accident detection, proper GPS location tracking, and reliable wireless alert transmission.

4.2 Simulation

The simulation of the Accident Detection and Alert System was carried out using Proteus simulation software and Embedded C programming in MPLAB X IDE. The purpose of simulation was to verify the logic and communication between the modules before hardware implementation.

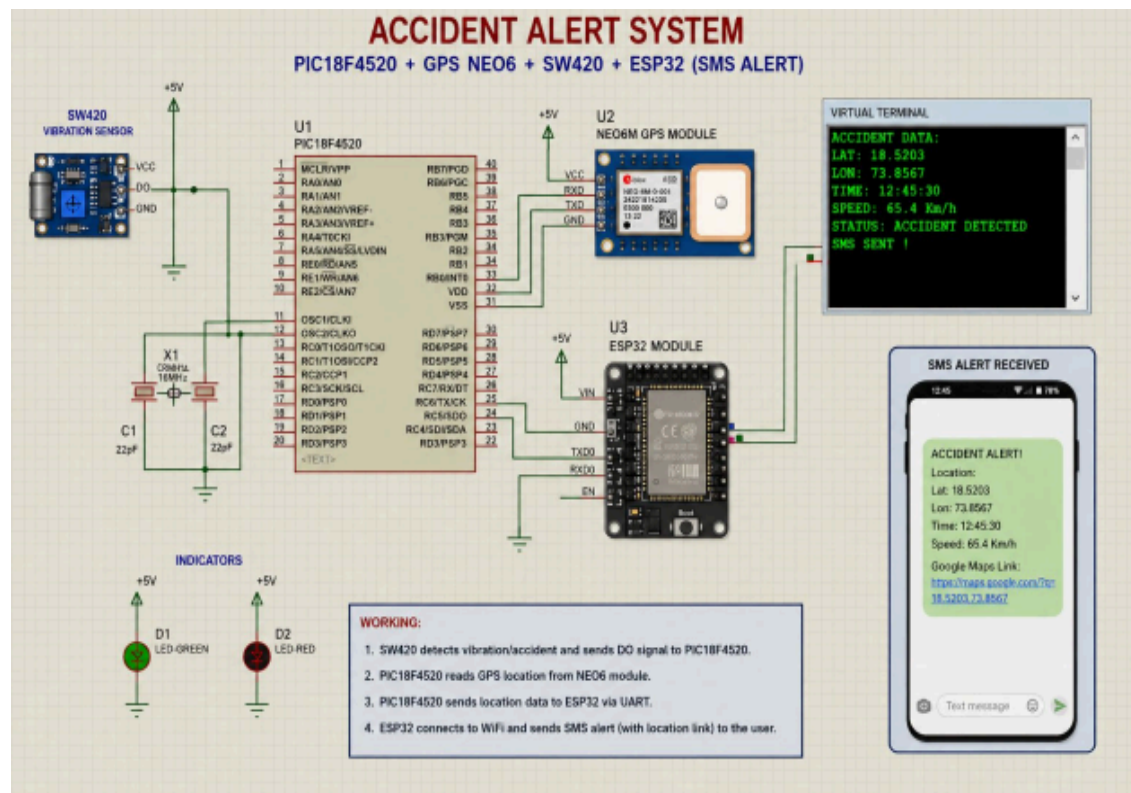
Simulation Process

- The vibration sensor output was simulated as an input signal to the PIC microcontroller.
- The PIC controller processed the vibration data and detected accident conditions based on threshold values.
- GPS module communication was simulated using serial UART communication.
- The ESP32 communication process was verified for transmitting alert messages.
- LCD outputs and status indications were monitored during testing.

Simulation Results

- Successful detection of abnormal vibration conditions.
- Correct retrieval of GPS coordinates.
- Proper communication between PIC microcontroller and ESP32.
- Accurate display of system status and alert generation.

The simulation helped reduce hardware errors and improved overall system reliability before prototype development.



4.3 PCB prototype:

After successful simulation, a PCB prototype of the Accident Detection and Alert System was developed for real-time implementation. The PCB was designed to integrate all hardware components in a compact and organized manner.

Prototype Development

- The circuit was first assembled on a breadboard for initial testing.
- PCB layout design was created using PCB design software.
- Components were soldered onto the PCB board carefully.
- Power and communication lines were tested for continuity and stability.

Prototype Testing

- Accident detection response was tested using vibration inputs.
- GPS module successfully generated real-time coordinates.
- ESP32 transmitted alert data successfully through wireless communication.
- The prototype operated reliably under different testing conditions.

The PCB prototype improved system stability, portability, and overall performance.

4.4 Mobile / Laptop Application (Blynk Interface)

A mobile or laptop monitoring interface can be used to receive accident alert notifications and location details transmitted by the ESP32 module. The application provides real-time monitoring and helps emergency contacts access accident information quickly.

4.5 Location Tracking (GPS Output)

The figure shows real-time location tracking using the NEO-6M GPS Module. The GPS data (latitude and longitude) is processed by the ESP32 and sent to the Blynk. The location is displayed on Google Maps through a link, showing the

exact position of the user. This helps in quick tracking during emergency situations.

5. RESULTS & PHOTOGRAPHS

5.1 Introduction:

The results and testing phase of the Accident Detection and Alert System was carried out to evaluate the performance and reliability of the developed prototype. The system was tested under different vibration conditions to verify accident detection accuracy, GPS location tracking, and wireless alert transmission.

The developed system successfully integrated the PIC18F4520 microcontroller, Vibration Sensor, GPS module, and ESP32 into a functional accident monitoring system. Experimental testing confirmed that the system can detect sudden impacts and immediately generate alert notifications with location details.

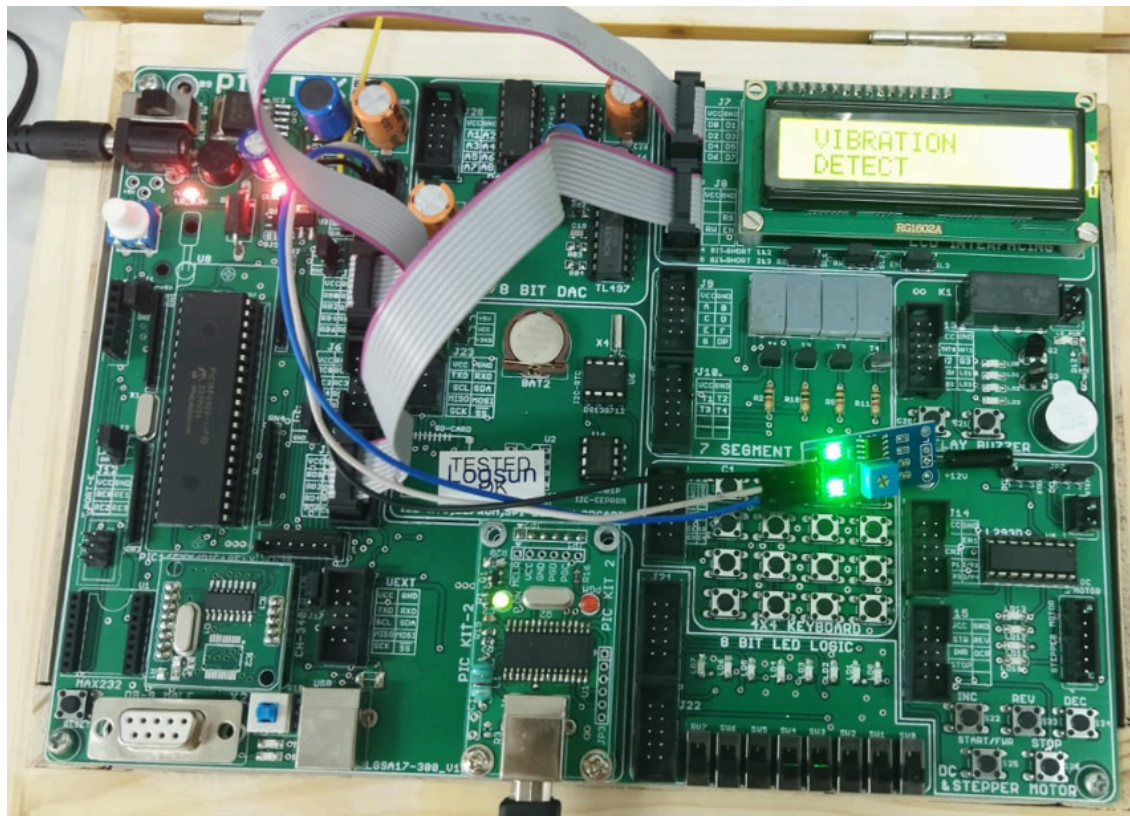
Photographs of the hardware setup, PCB prototype, simulation results, and testing process can be included in this section to demonstrate the practical implementation of the project.

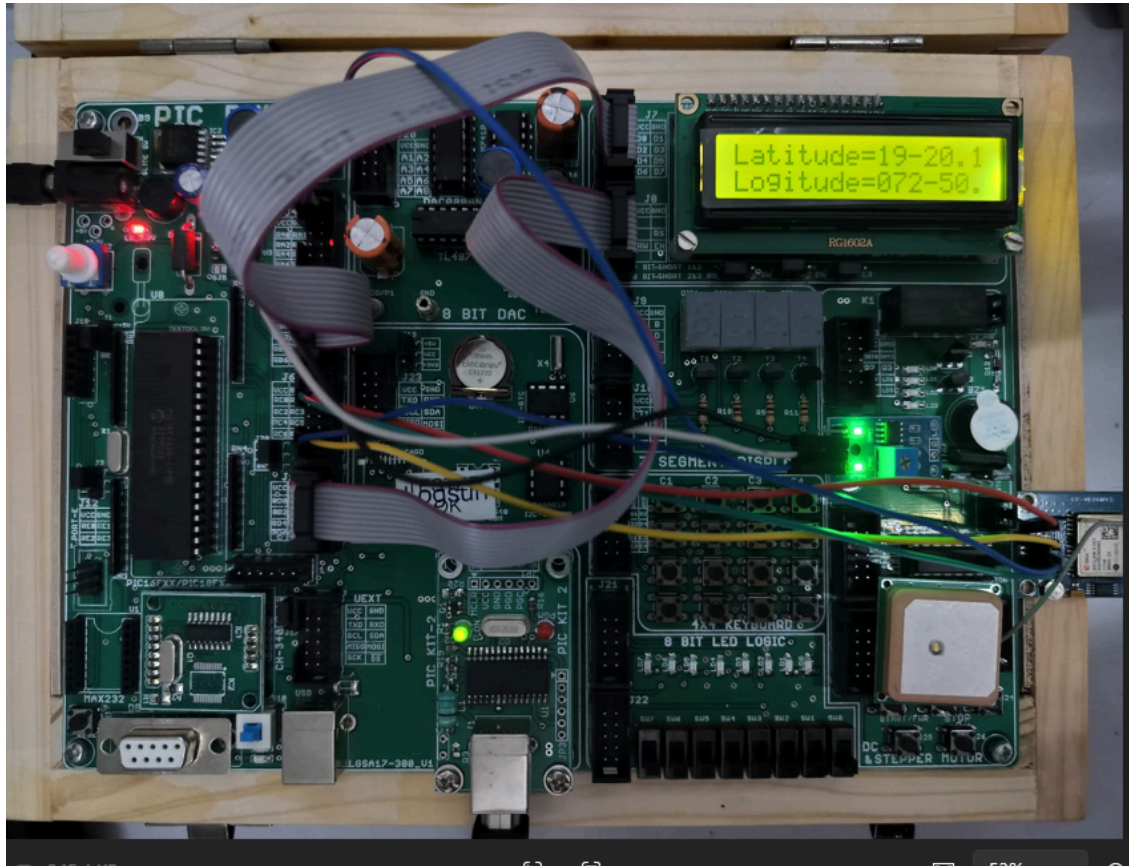
5.2 Results:

- The vibration sensor successfully detected accident vibrations and shocks.
- The PIC18F4520 accurately processed sensor data and detected accident conditions.

- The GPS module successfully provided real-time location coordinates.
- The ESP32 transmitted accident alert messages through Wi-Fi successfully.
- The system responded quickly with reliable performance during testing.
- The developed prototype operated efficiently with low power consumption.
- The Accident Detection and Alert System successfully achieved automatic accident monitoring and alert generation.

5.3 Photographs:





6. CONCLUSIONS AND FUTURE SCOPE

6.1 Conclusion:

The Accident Detection and Alert System was successfully designed and implemented using the PIC18F4520, Vibration Sensor, GPS module, and ESP32. The system effectively detects accident vibrations and automatically sends alert information with the accident location.

The developed prototype demonstrated reliable performance in detecting sudden impacts and transmitting emergency notifications. The system helps reduce emergency response time and improves vehicle safety through automatic accident monitoring and communication.

6.2 Future Scope

- Integration of GSM module for SMS-based emergency alerts
- Development of a mobile application for real-time monitoring
- Cloud storage for accident data and tracking history
- AI-based accident severity analysis
- Addition of camera module for live accident monitoring
- Integration with ambulance and emergency service systems
- Use of IoT platforms for smart vehicle monitoring
- Improvement in GPS accuracy and communication speed

7.1 References:

Research Paper :

- Heart Rate Sensor with IoT Features.

https://www.researchgate.net/publication/379859513_Heart_rate_sensor_with_IoT_features

- Portable IoT-Based Health Monitoring Framework (2025).

https://www.researchgate.net/publication/392123135_A_Portable_IoT-Based_Health_Monitoring_Framework_Using_ESP32_for_Isolated_and_Home-Based_Patient_Care.com

- MAX30102 + ESP32 for Oxygen & Heart Rate Monitoring.

<https://www.mdpi.com/2673-4591/16/1/9.com>

